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Variable relationship discovery and recommendations for industrial automation environments

Abstract

Various embodiments of the present technology generally relate to solutions for improving industrial automation programming and data science capabilities with machine learning. More specifically, embodiments of the present technology include systems and methods for implementing machine learning engines within industrial programming and data science environments to improve performance, increase productivity, and add functionality. In an embodiment, a system comprises a machine learning-based analysis engine configured to identify a variable that is available to be utilized in control logic for controlling an industrial automation environment. The machine learning-based analysis engine is further configured to determine that the variable is not utilized in the control logic. A recommendation component of the system is configured to, in an industrial programming environment, surface a recommendation to add the variable to the control logic.

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Background/Summary

BACKGROUND

(1) Industrial manufacturing environments generate huge quantities of data at very fast speeds making the extraction of enterprise-level insights challenging. In industrial automation environments, control systems are used to drive various operations along an industrial line. Control code is used by industrial drives or programmable logic controllers to drive industrial assets, devices, and sensors in an industrial process. Operational data produced during runtime contains important information about inputs, outputs, operations, status, performance, or quality of the industrial process, but can be difficult to leverage in control code programming given the enormous amount of computing power and time that goes into data science and data analytics. Moreover, control programs are typically developed by programmers without access to or with limited access to runtime data, statistics, or the like. Important integrations, dependencies, variables, inputs, or outputs can be easily overlooked by control programmers because of the lack of accessibility to useful operational information. Moreover, an industrial process may have thousands of relevant variables, making it difficult for programmers to adequately provide connections and logic for every important variable, tag, or device. Manually editing control programs in response to various information in operational data can be an extremely difficult and time-consuming process that requires intimate knowledge of data science, data analytics, and process control. Data scientists face similar challenges in that they may be provided enormous amounts of operational data from one or more industrial automation environments with which they may wish to perform data mining, data processing, predictive modeling, or visualization, but the data scientist likely has no insight into the control logic or the industrial automation environment itself.

(2) Machine learning algorithms are designed to recognize patterns and automatically improve through training and the use of data. Many different types of machine learning models exist including classification models, regression models, clustering models, dimensionality reduction models, and deep learning models. Examples of machine learning algorithms include artificial neural networks, nearest neighbor methods, decision trees, ensemble random forests, support vector machines (SVMs), naïve Bayes methods, regressions, and more. A machine learning algorithm comprises an input layer and an output layer, wherein complex analyzation takes places between the two layers. Various training methods are used to train machine learning algorithms wherein an algorithm is continually updated and optimized until a satisfactory model is achieved. One advantage of machine learning algorithms is their ability to learn by example, rather than needing to be manually programmed to perform a task, especially when the tasks would require a near-impossible amount of programming to perform the operations in which they are used.

(3) It is with respect to this general technical environment that aspects of the present disclosure have been contemplated. Furthermore, although a general environment is discussed, it should be understood that the described examples should not be limited to the general environment identified in the background.

Overview

(4) This Overview is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(5) Various embodiments of the present technology generally relate to solutions for improving industrial automation programming and data science capabilities with machine learning. More specifically, embodiments of the present technology include systems and methods for implementing machine learning engines within industrial programming and data science environments to improve performance, increase productivity, and add functionality. In an embodiment of the present technology, a system comprises a memory that stores executable components and a processor, operatively coupled to the memory, that executes the executable

components. The executable components comprise a machine learning-based analysis engine configured to identify a variable that is available to be utilized in control logic for controlling an industrial automation environment. The machine learning-based analysis engine is further configured to determine that the variable is not utilized in the control logic. The executable components further comprise a recommendation component configured to, in an industrial programming environment, surface a recommendation to add the variable to the control logic.

(6) In an embodiment of the present technology, the executable components further comprise a programming component configured to, in response to an input accepting the recommendation to add the variable to the control logic, add the variable to the control logic. The executable components may further comprise a determination component configured to determine that the recommendation to add the variable to the control logic should be surfaced based on a frequency of utilization of the variable in control logic for controlling industrial automation environments of the same type as the industrial automation environment. In some examples, the frequency of utilization of the variable is determined based on information derived from one or more control code repositories to which the system is communicatively coupled. The machine learning-based analysis engine, to identify the variable, may analyze operational data from the industrial automation environment. The machine learning-based analysis engine, to identify the variable, may also search for variables that are correlative to changes in outcome in the operational data. In an exemplary embodiment, determining that the variable is not utilized in the control logic comprises receiving an output from a machine learning model of the machine learning-based analysis engine.

(7) In another embodiment, a non-transitory computer-readable medium has stored thereon instructions that, in response to execution, cause a system comprising a processor to perform operations. The operations comprise identifying, in a machine learning-based analysis engine, a variable that is available to be utilized in control logic for controlling an industrial automation environment. The operations further comprise determining, in the machine learning-based analysis engine, that the variable is not utilized in the control logic and surfacing a recommendation, in an industrial programming environment, to add the variable to the control logic.

(8) In yet another embodiment, a method of recommending the integration of salient variables in an industrial automation programming environment comprises identifying, by a system comprising a processor, a variable that is available to be utilized in control logic for controlling an industrial automation environment. The method further comprises determining, by the system, that the variable is not utilized in the control logic and surfacing, by the system and in an industrial programming environment, a recommendation to add the variable to the control logic.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views. While several embodiments are described in connection with these drawings, the disclosure is not limited to the embodiments disclosed herein. On the contrary, the intent is to cover all alternatives, modifications, and equivalents.

(2) FIG. 1 illustrates an overview of an industrial automation environment in which aspects of the present technology are implemented;

(3) FIG. 2 illustrates a user interface environment for programming control logic in accordance with some embodiments of the present technology;

(4) FIG. 3 illustrates an example of an industrial automation environment in which some embodiments of the present technology may be implemented;

- (5) FIG. 4 illustrates a user interface environment for programming control logic in accordance with some embodiments of the present technology;
- (6) FIG. 5 illustrates a user interface environment for programming control logic in accordance with some embodiments of the present technology;
- (7) FIG. 6 illustrates a user interface environment for programming control logic in accordance with some embodiments of the present technology;
- (8) FIG. 7 illustrates a user interface environment for programming control logic in accordance with some embodiments of the present technology;
- (9) FIG. 8 illustrates a series of steps for utilizing machine learning in industrial automation programming environments in accordance with some embodiments of the present technology;
- (10) FIG. 9 illustrates a series of steps for utilizing machine learning in industrial automation programming environments in accordance with some embodiments of the present technology;
- (11) FIG. 10 illustrates a series of steps for utilizing machine learning in industrial automation programming environments in accordance with some embodiments of the present technology;
- (12) FIG. 11 illustrates a series of steps for utilizing machine learning in industrial automation programming environments in accordance with some embodiments of the present technology;
- (13) FIG. 12 illustrates a series of steps for utilizing machine learning in industrial automation data science environments in accordance with some embodiments of the present technology; and
- (14) FIG. 13 illustrates an example of a computing device that may be used in accordance with some embodiments of the present technology.
- (15) The drawings have not necessarily been drawn to scale. Similarly, some components or operations may not be separated into different blocks or combined into a single block for the purposes of discussion of some of the embodiments of the present technology. Moreover, while the technology is amendable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the technology to the particular embodiments described. On the contrary, the technology is intended to cover all modifications, equivalents, and alternatives falling within the scope of the technology as defined by the appended claims.

DETAILED DESCRIPTION

- (16) The following description and associated figures teach the best mode of the invention. For the purpose of teaching inventive principles, some conventional aspects of the best mode may be simplified or omitted. The following claims specify the scope of the invention. Note that some aspects of the best mode may not fall within the scope of the invention as specified by the claims. Thus, those skilled in the art will appreciate variations from the best mode that fall within the scope of the invention. Those skilled in the art will appreciate that the features described below can be combined in various ways to form multiple variations of the invention. As a result, the invention is not limited to the specific examples described below, but only by the claims and their equivalents.
- (17) Various embodiments of the present technology generally relate to solutions for improving industrial automation programming and data science capabilities with machine learning. More specifically, embodiments of the present technology include systems and methods for implementing machine learning engines within industrial programming and data science environments to improve performance, increase productivity, and add functionality. Generally, industrial control logic provides instructions for controlling and performing certain operations within an industrial environment via controllers (e.g., programmable logic controllers or “PLCs”), wherein the controllers execute the control code to control downstream devices and machinery. Engineers and/or programmers are generally responsible for generating the control logic used to drive industrial processes. Data scientists are generally responsible for analyzing and processing operational data produced by the industrial automation environment to find important correlations, build models, create visualizations, and generally produce information that may be useful at the enterprise level, control level, or the field level. The present technology serves to bring these

environments together such that engineers may utilize operational data in a useful manner to improve control logic programming and data scientists may utilize control logic and modeling data to improve their analytic capabilities.

(18) Thus, the present technology serves to greatly enhance engineering and data science in industrial automation contexts by integrating their environments along with machine learning-based tools for assisted programming and assisted data analytics. In some embodiments, a machine learning-assisted programming environment is provided such that an engineer or programmer may be presented with recommended components, connections, and the like as they create code. In some examples, a machine learning-based engine (i.e., a “wizard”) may suggest automatic completions (“auto-completions”) of code, wherein a suggested auto-completion may include one or more components and connections. Suggested auto-completions may also include configurations or settings for components of the control code. A machine learning model, in an embodiment, consumes data from existing control code in the programming environment, or the integrated development environment (IDE), to create suggestions or automatic completions. As an engineer writes control logic, the auto-complete program may suggest useful or relevant tags or surface relevant data in the context of the controlled process. The auto-complete program, in some embodiments, uses historical operational data to inform suggestions it makes in the programming environment.

(19) In addition to suggestions and auto-completions, the programming environment, in an embodiment, provides access to operational data as well as data science tools (e.g., data analysis tools, data presentation tools, data mining tools, modeling tools, etc.) for the user and/or the machine learning engine to use during a programming session. Operational data, in some examples, includes data from the industrial automation environment that the control code is used in. In other examples, the operational data includes data from other environments that may be similar to the environment in which the control code will be implemented.

(20) Moreover, in addition to operational data and data science tools being accessible from within the automation engineering environment or IDE, the present technology provides for automation engineering tools and data to be accessible from within a data science environment. A data science environment, as described herein, may include any computing environment in which data scientist functions, data analyst functions, data architect functions, statistician functions, database administrator functions, business analyst functions, or data and analytics manager functions may be performed. Such functions may include but are not limited to data mining, data modeling, predictive modeling, visualization, statistics, machine learning development, spreadsheet tools, systems development, enterprise resource planning, and business or enterprise-level intelligence. In an example, a data scientist may be presented with operational data collected from one or more industrial automation sites in an analytics environment while trying to mine for correlations and patterns. However, the data provides little to no information or context about the control logic, models, assets, or the like that produced the data. Thus, the data scientist may, in an example, select a piece of data and be presented with information, diagrams, or models relevant to the data. For example, a control diagram for a region proximate or in connection with a component that produced the particular chunk of data may be surfaced. In another example, a pipeline of data feeding into or out of the particular chunk of data may be surfaced. Other contextual information relevant to data that is typically available in a data science environment may be similarly surfaced, either through selection of a piece of data, selection of a menu that may provide contextual data, automatically surfaced, or the like. In an exemplary embodiment, surfacing relevant contextual data is performed in part by a machine learning-based context engine that identifies what contextual data is relevant to the particular chunk of data.

(21) In accordance with the present disclosure, an industrial asset library and wizard are provided in an industrial programming and/or automation engineering environment. An industrial asset library, in an example, provides a list of industrial assets including but not limited to models,

controllers, sensors, actuators, and other devices and allows a user to choose and configure an asset. In some examples, the asset library presents for a user a list of only relevant assets that may be useful based on the control logic or target industrial environment, wherein determining which assets to include in the asset library is achieved via a machine learning engine trained to identify relevant assets based on the industrial environment and/or control logic. In another example, the industrial asset library is accessible to and used by a machine learning-based recommendation engine (i.e., the “wizard”) such that the recommendation engine can generate recommendations or auto-completions for the control logic.

(22) In some examples, the industrial asset library includes a library of machine learning models specific to industrial automation applications, wherein the library of machine learning models is accessible to automation engineers and the wizard. The library, in an embodiment, bucketizes inputs based on categories. As an example, categories that may be used to bucketize inputs include but are not limited to tag data, physical input devices, environmental inputs, economic or other macro-environment inputs, and cyclical or seasonal inputs. To provide relevant inputs, arrays of different assets and/or characteristics may be utilized to assist automation engineers in finding valuable inputs to include in their logic.

(23) The wizard, in accordance with the present technology, helps integrate models, variables, inputs, tags, and the like into the control code. The wizard is a machine learning-backed recommendation and configuration engine that helps choose models, variables, inputs, tags, and the like as well as provides assistance with tag linking and input/output (I/O) linking. The one or more machine learning models of the wizard are streamed data to generate its predications, wherein the data may come from many different places including the live industrial environment or network databases. In one example, the wizard may provide a user with a “short list” of variables that the engineer may want to control, wherein the short list is a subset of a full list of available variables.

(24) In an example, an engineer or programmer may add an asset to the programming code in the programming environment. The wizard may then provide a list of tags that are related to the asset, available for linking to the asset, or commonly linked to the asset. Similarly, the engineer may choose to open a truncated list of tags from the library without the use of the recommendation wizard. The machine learning engine may, based on a starting set of devices, provide the engineer with the standard set of devices for a particular type of asset in the code. In another example, an engineer may be building a pipeline for a machine learning model they are implementing in the control code. In this example the wizard or asset library may be used to pull together the assets that should be used to make the machine learning model operable. The library or wizard may also suggest environmental factors, such as ambient air temperature, that are impactful in the data pipeline. Moreover, the wizard may be used to alert an engineer of any inconsistencies in the control code that may prevent it from functioning properly.

(25) In addition to the wizard and library, the present disclosure contemplates the addition of other data science tools into the IDE or programming environment. For example, operational data and industrial environment data may be accessible from within the IDE. In an example, an automation engineer using the IDE may be building control logic for a production line. While the programming environment provides information about the flow of the process, it also provides access to the historical data stream for the production line, such as the data for each tag and process object in the line. The engineer may select a tag, and a machine learning-based engine may identify inputs and tags upstream of the selected tag that likely have an influence on that tag. In this way, the engineer can rapidly find relevant data in the context of the process from within their IDE. The engineer can then assess the relevance of data points to what they are trying to improve.

(26) The present disclosure also contemplates the flip side of the previously discussed technology: the inclusion of contextual information from the programming and/or industrial environments in the data science environment. By integrating information from the programming and/or industrial environments into the data science environment, a data scientist may have far more contextual

information for the data they are working with than they traditionally would. In this enhanced data science environment, connections may exist between the data science environment and the operational environment. In an exemplary embodiment, the data science environment includes an application programming interface (API) into the live environment, providing contextual information about a selected tag or column of data in some examples.

(27) A data scientist, in an example, may receive a dump of data from which they seek to find relationships, correlations, patterns, and the like that can be used later to design models. A data scientist may wish to determine which tags, datapoints, sensors, and instrumentation are relevant to solve a problem based on the statistical analysis of data. The lack of contextual information, however, may severely limit the ability of a data scientist to find important data and design effective models. Thus, integrating the data scientist's environment with the programming environment, along with machine learning-based programs, allows hints and other contextual information to be surfaced for a data scientist. Contextualizing the data with specific process steps, assets, equipment, I/O nodes, and the like allows a data scientist to accelerate the feature selection process and identify which datapoints are relevant to a particular problem. This allows for a more streamlined and targeted data science process than mining for correlations in a non-contextualized dataset. In another example, statistics from sensor data may indicate that there is a heavy reliance on one I/O node versus another. Surfacing such information about the different nodes can improve understanding and awareness of the automation process by the data scientist.

(28) In some embodiments, the programming environments and data science environments previously described are integrated into a singular, multi-purposeful environment such that the engineering and data science disciplines can effectively collaborate. In this way, a process engineer who is defining control logic for a line or asset, can leverage data science capabilities or collaborate with a data scientist to program the control logic in a more robust manner.

(29) Embodiments of the described technology may be implemented to perform variable relationship discovery. Variable relationship discovery may include finding correlations between variables, which is traditionally a manual process in which engineers use their own judgment to select variables that they think should be included in the process. However, when shifting to big data, it becomes far more difficult to identify how variables are related or correlated. Thus, in accordance with the present disclosure, machine learning models are used within the programming environment to help find potentially useful variables, models, or components for an industrial line.

(30) A relationship discovery engine comprising one or more machine learning models is used to analyze control code and consume data to identify which variables have a large impact on the process and important variable relationships. The data consumed may, in some embodiments, include operational data. A machine learning-based engine may be used to identify a variable that is heavily used in a particular industrial process but is not being used or is barely used in the control code. The programming environment thus notifies a user that the variable has been potentially overlooked and/or recommends the addition of the variable.

(31) In an example, a machine learning-based engine is used to determine that humidity is predictive of temperature increases in the industrial automation environment, but that a humidity variable is not included in the control code. Thus, a recommendation to add the humidity variable to the control logic is surfaced. In response to an acceptance of the recommendation, the variable may be added to the control logic and configured. In another example, variable relationship discovery is performed with respect to big data mining and data repositories. In such an example, a machine learning-based engine analyzes a code repository to determine dominant variables in a codebase. In certain implementations, the engine may look at various types of data available in the code base, including, in some instances, a comment section. Thus, when an engineer is programming an industrial line with the assistance of a machine learning engine, the engine may recognize that there are 100 tags available and three of them have not been used anywhere in the code. The engine further recognizes that those three missing tags are used in many other code

repositories and informs the user of how the tags are typically used.

(32) In some examples the machine learning-based engine is used to recommend the insertion of a loop, a function, a variable, or the like as well as input to the component and assistance with configuring and/or connecting the component in the control code. Conversely, a machine learning-based engine may identify that a variable that is heavily used in the control code is absent or under-used in the data pipeline, suggesting that its inclusion in the pipeline and model may yield better predictions.

(33) Embodiments of the described technology may be further implemented to perform variable slimming. If, during runtime, extra variables are being output by the control logic that are not needed, computing power and storage are wasted. Data pipelines that are over-inclusive waste storage space and processing power, and extraneous data may reduce the predictive capability of some models. Thus, a machine learning-based engine may be used to inform, during a programming session, that a variable that is in the control program is not commonly used, isn't generally useful, or isn't useful in the particular industrial line based on historical process data. The engine may therefore recommend removing the variable from the control logic or automatically analyze datasets for extraneous data and eliminate the data from the pipeline.

(34) In accordance with the present disclosure, a machine learning model comprises one or more machine learning algorithms that are trained based on historical data and/or training data. A machine learning model may employ one or more machine learning algorithms through which data can be analyzed to identify patterns, make decisions, make predictions, or similarly produce output that can inform control code and/or parameters. Different types of machine learning models may be used in accordance with the present technology including but not limited to classification models, regression models, clustering models, dimensionality reduction models, and deep learning models. Examples of machine learning algorithms that may be employed solely or in conjunction with one another include artificial neural networks, nearest neighbor methods, decision trees, ensemble random forests, support vector machines (SVMs), naïve Bayes methods, regressions, or similar machine learning techniques or combinations thereof capable of predicting output based on input data. Determining which machine learning methods to use may depend on the specific purpose or functions required for a particular industrial setting, the specific programming application used, or similar requirements. A machine learning engine, in some examples, outputs a prediction about what components might be beneficial to add or remove from the control logic. Similarly, a machine learning engine may output a recommendation to add or remove one or more components from the control logic, or a notification that a component that should be used or is typically used is missing. In other examples the machine learning engine outputs the predicted components or settings directly for integration into the control code. In yet another example, a machine learning engine provides a model of the control code or the industrial automation environment in such a way that a data scientist may use the contextual information to inform their analyses. Other outputs with a similar purpose may exist and are contemplated herein.

(35) FIG. 1 illustrates an overview of an industrial automation environment wherein engineering and data science environments are integrated as described herein. FIG. 1 includes automation environment **100**. Automation environment **100** comprises physical system **101** including controlled process **102**. Automation environment **100** further comprises digital twin **103** and modeling environment **110**. Modeling environment **110** comprises historical observations **111**, machine learning **112**, instructions and first principles **113**, and control logic **114**. Automation environment **100** further comprises data scientist **121**, engineer **122**, model **130**, controller **140**, actuators **150**, and sensors **160**.

(36) In the example of FIG. 1, digital twin **103** of physical system **101** is provided to modeling environment **110**, wherein modeling environment **110** is used by both data scientist **121** and engineer **122**. In one embodiment, data scientist **121** accessed modeling environment **110** through a different application than engineer **122**. In an alternative embodiment, data scientist **121** and

engineer **122** use the same application for their data science and engineering purposes. In either embodiment, information from the data science portion of the modeling environment and the engineering portion of the modeling environment is shared and accessible for both functions—data scientist **121** can access contextual engineering data in performing data science tasks and engineer **122** can access operational data and data science tools. As shown, a machine learning-based engine is provided in both the engineering environment and the data science environment. Functionality of the machine learning-based engines may include generating recommendations, auto-completion functionality, providing contextual information, and other functionalities described herein.

Instructions and first principles **113** is used to generate control logic **114**. Engineer **122** may utilize operational data or similar from the data science environment in producing both instructions and first principles **113** and control logic **114**. In some examples, a machine learning-based recommendation engine recommends adding components or editing control logic based in part on historical observations **111** and machine learning **112**.

(37) Model **130** is representative of one or more machine learning models implemented in control code for controlling controlled process **102**. The one or more machine learning models may include predictive models for generating forward inferences and diagnostic models for identifying causal relationships. Model **130** includes one or more machine learning models that produces control logic as output for controller **140**. Controller **140** runs one or more prescriptive models for optimizing controlled process **102** or a portion of controlled process **102**. Controller **140** controls actuators **150** which actuate physical system **101** to perform controlled process **102**. Actuators **150** comprise one or more machine learning models for autonomous driving of physical system **101**. Sensors **160** collect data from physical system **101** running controlled process **102**. Sensors **160** comprise one or more machine learning models for describing output and current state from controlled process **102**. Output from controlled process **102** is also provided to modeling environment **110** and may be used for data science and/or engineering purposes.

(38) FIG. 2 illustrates a graphical user interface (GUI) of an industrial automation programming environment in accordance with some embodiments of the present technology. FIG. 2 comprises GUI **200**, which is representative of any user interface for writing and editing industrial control code. In some examples, GUI **200** is representative of modeling environment **110** from FIG. 1. GUI **200** includes workspace **210** for writing and editing control code. Workspace **210** includes control logic **220**, a current project for control code for an industrial automation process. GUI **200** also includes wizard **230**. In the present example, wizard **230** has recommended adding component **231**, component **232**, link **233**, component **234**, link **235**, and link **236** to control logic **220**.

(39) Wizard **230** comprises one or more machine learning models for predicting and/or recommending components for control logic. Wizard **230** may be configured to auto-complete all or a portion of existing control logic in workspace **210**. Wizard **230** may alternatively be configured to surface recommendations to add components to existing control logic in workspace **210**, wherein the recommendations may be verbal or may display the recommended additions themselves as is shown in the example of FIG. 2. In some examples a user of the engineering environment shown in FIG. 2 selects wizard **230** to trigger it to generate and show recommendations. In other examples, wizard **230** is displayed just to show that the wizard is operating and will automatically surface recommendations as they are generated.

(40) FIG. 3 illustrates an example of an industrial automation environment that may be representative of industrial automation environments as discussed herein. The industrial automation environment of FIG. 3 comprises enterprise environment **310**, control environment **320**, and field environment **330**. The industrial automation environment of FIG. 3 may include fewer or additional sub-environments than those shown. Likewise, the sub-environments of the industrial automation environment of FIG. 3 may include fewer or additional components, assets, or connections than shown.

(41) Enterprise environment **310** comprises engineering and design environment **311**, visualization

environment **312**, and information environment **313**. Control environment **320** comprises database server **321**, management servers **322**, web server **323**, scheduling server **324**, and application server **325**. Each of database server **321**, management servers **322**, web server **323**, scheduling server **324**, and application server **325** may be representative of a single server or a plurality of servers. Field environment **330** comprises PLC **331**, which is coupled to human-machine interface (HMI) **332** and machine **133**. Field environment **330** also comprises PLC **334**, which is coupled to machine **335**. Field environment **330** further comprises PLC **336**, which is coupled to HMI **373** and machine **338**. Field environment **330** also comprises machine learning asset (MLA) **339**, which is coupled to HMI **340**.

(42) In some examples, engineering and design environment **311** is representative of any connected devices or environments on which control logic is programmed for operating devices in field environment **330**. Visualization environment **312** is representative of any connected devices or environments on which operational data is viewed and/or analyzed. Information environment **313** is representative of any connected devices or environments on the enterprise level on which any other industrial automation information relevant to field environment **330** can be viewed, processed, analyzed, generated, or the like. Assets of control environment **320** are illustrative in nature such that one, none, or any combination of known server types may be in communication with the industrial automation environment of FIG. **3**. Similarly, field environment **330** is illustrative in nature such that one, none, or any combination of assets may be present.

(43) Each of PLC **331**, PLC **334**, and PLC **336** is representative of one or more programmable logic controllers, which may be coupled to one or more devices, machines, sensors, actuators, interfaces, or other asset types. Each of machine **333**, machine **335**, and machine **338** is representative of any industrial machine relevant to the industrial automation environment of FIG. **3**. Each of HMI **332**, HMI **337**, and HMI **340** is representative of any form of human-machine interface on which a user or operator in field environment **330** can view and/or interact with connected assets.

(44) Machine learning asset **339** is representative of any machine learning model implemented within the industrial automation environment of FIG. **3** as described herein. As previously discussed, machine learning asset **339** may take input from field environment **330** or external sources, such as any server in control environment **320**, to generate an output useful to controlling any processes or assets in field environment **330**. Machine learning asset **339** may be implemented in a fully closed-loop process or may receive external data for use in generating predictions and/or outputs.

(45) FIG. **4** illustrates an example of a programming environment in which machine learning assets can be programmed into an industrial process. FIG. **4** includes user interface **400**, which is representative of any user interface for generating control code. In some examples, user interface **400** is representative of modeling environment **110** from FIG. **1**. In some examples, user interface **400** is running in engineering and design environment **311** from FIG. **3**. User interface **400** includes explorer **410**, wherein explorer **410** comprises industrial asset libraries, the industrial asset libraries including model library **411**, programmable assets library **412**, and tags library **413**. User interface **400** includes workspace **440** comprising industrial line **420**. Industrial line **420** includes database **421**, external variable **422**, machine learning asset **423**, HMI **424**, PLC **425**, HMI **426**, machine **427**, temperature sensor **428**, and speed sensor **429**. User interface **400** further includes wizard **430** and suggestion **431**.

(46) In industrial line **420**, database **421** and external variable **422** provide input to machine learning asset **423**. Database **421** may comprise a model database that is operatively coupled to the industrial line such that machine learning asset **423** can be swapped with other models stored in database **421**. External variable **422** may be representative of any external information that may be used as input to machine learning asset **423**. Machine learning asset **423** is coupled to HMI **424**, which displays information relevant to the status or control of machine learning asset **423**. Machine

learning asset **423** is coupled to PLC **425**, which is coupled to HMI **426**. HMI **426** displays information relevant to the status or control of PLC **425**. Machine **427** is coupled to and driven by PLC **425**. Temperature sensor **428** and speed sensor **429** measure temperature and speed associated with machine **427**.

(47) A programmer may develop or edit industrial line **420** within user interface **400** in accordance with the present disclosure. A programmer has access to model library **411** comprising machine learning models that can be used in industrial line **420**. A programmer also has access to programmable assets library **412** comprising programmable assets that can be used in industrial line **420**. A programmer also has access to tags library **413** comprising tags that can be used in industrial line **420**. In some examples, assets may be provided such that a programmer can “drag and drop” assets into their project such as industrial line **420** and connect assets to the existing model in the same way that other assets are typically connected in the programming environment. In some examples, the asset library presents for a user only a list of relevant assets that may be useful based on the control logic or target industrial environment, wherein determining which assets to show in the asset library is achieved via a machine learning engine trained to identify relevant assets based on the industrial environment and/or control logic. In another example, the industrial asset library is accessible to and used by wizard **430** such that the recommendation engine can generate recommendations or auto-completions for the control logic.

(48) Wizard **430** helps integrate models, variables, inputs, tags, and the like into industrial line **420**. Wizard **430** is a machine learning-based recommendation and configuration engine that helps choose models, variables, inputs, tags, and the like as well as provides assistance with tag linking and I/O linking. Wizard **430** comprises one or more machine learning models. The one or more machine learning models of the wizard are streamed data to generate predications, wherein the data may come from many different places including the live industrial environment or network databases. In one example, the wizard may provide a user with a “short list” of assets that the engineer may want to add, wherein the short list is a subset of a full list of available assets.

(49) In the present example, wizard **430** has produced suggestion **431**, wherein suggestion **431** suggests adding PLC_2 to industrial line **420**. In an example, an engineer or programmer may accept the recommendation to add an asset to industrial line **420** in the programming environment. Wizard **430** may then provide a list of tags that are related to the asset, available for linking to the asset, or commonly linked to the asset. Similarly, the engineer may choose to open a truncated list of tags from the library without the use of wizard **430**.

(50) FIG. 5 illustrates a user interface of an industrial automation programming environment for performing variable relationship discovery in accordance with some embodiments of the present technology. FIG. 5 comprises GUI **500**, which is representative of any user interface for writing and editing industrial control code. In some examples, GUI **500** is representative of modeling environment **110** from FIG. 1. In some examples, GUI **500** may be running in engineering and design environment **311** from FIG. 3. GUI **500** includes workspace **520** for writing and editing control code. Workspace **520** includes control logic **510**. GUI **500** also includes wizard **530** and recommendation **531**. In the present example, wizard **230** has surfaced recommendations to add component **532**, component **533**, link **534**, component **535**, and link **536** to control logic **510**. Wizard **530** has also surfaced a recommendation to add a tag in recommendation **531**.

(51) Wizard **530** is representative of any machine learning-based engine capable of recommending industrial components for control code. In some examples wizard **230** consumes data from existing control code (i.e., control logic **510**) in the IDE to create suggestions and automatic completions as shown in the present example. As an automation engineer writes control logic, the wizard **530** may suggest useful or relevant tags or surface relevant data in the context of control logic **510**. Wizard **530**, in some embodiments, uses historical operational data to inform suggestions it makes in the IDE.

(52) Variable relationship discovery, in accordance with the present example, includes the use of

machine learning models (i.e., those within wizard **530**) for finding correlations between variables to help discover potentially useful variables, models, or components for an industrial line. Wizard **530** may serve as a relationship discovery engine comprising one or more machine learning models used to analyze control logic **510** and consume data to identify which variables have a large impact on the process and important variable relationships. The data consumed may, in some embodiments, include operational data. Wizard **530** may be used to identify a variable that is heavily used in an associated industrial process but is not being used in control logic **510**. Wizard **530** thus surfaces recommendation **531** alerting the user that the variable has been potentially overlooked and recommends the inclusion of the variable.

(53) In some examples, wizard **530** is used to recommend the insertion of a loop, a function, a variable, or the like as well as input to the component and assist with configuring and/or connecting the component in the control code. Conversely wizard **530** may identify that a variable that is heavily used in the control code is absent or under-used in the data pipeline, suggesting that its inclusion in the pipeline and model may yield better predictions.

(54) FIG. **6** illustrates a user interface of an industrial automation programming environment for performing variable slimming in accordance with some embodiments of the present technology. FIG. **6** comprises GUI **600**, which is representative of any user interface for writing and editing industrial control code. In some examples, GUI **600** is representative of modeling environment **110** from FIG. **1**. In some examples, GUI **500** may be running in engineering and design environment **311** from FIG. **3**. GUI **600** includes workspace **610**, wherein workspace **610** comprises control logic **620** comprising tag **632**. GUI **600** further comprises wizard **630** and recommendation **631**. Wizard **630** is a machine learning-based engine comprising one or more machine learning models that may be used to inform, during a programming session, that a variable in the control program is not commonly used, is not generally useful, or is not useful in the particular industrial line based on historical process data. Wizard **630** may therefore recommend removing the variable from the control logic or automatically analyze datasets for extraneous data and eliminate the data from the pipeline.

(55) In the example of FIG. **6**, wizard **630** has recommended removing tag **632**, wherein the recommendation is expressed in a verbal format in recommendation **631** as well as with an indicator at tag **632** that it should be removed. In an example, a user of the programming environment may accept the recommendation to remove tag **632**. In some embodiments, acceptance of the recommendation may include manually removing tag **632**. In other embodiments, accepting the recommendation may include selecting a button to select the recommendation, at which time wizard **630** may remove tag **632** itself.

(56) FIG. **7** illustrates a user interface environment of an industrial automation programming environment integrated with an industrial data science environment. FIG. **7** includes GUI **700**. GUI **700** includes workspace **710** comprising control logic **720**, wherein control logic **720** comprises PLC **721**. GUI **700** further includes wizard **730**. GUI **700** further includes data table **740**, wherein data table **740** comprises operational data related to PLC **721** of control logic **720**. A user of the programming environment shown in FIG. **7** has used the “data access” menu to access operational data from PLC **721**. However, in other embodiments, operational data may be surfaced in response to selection of PLC **721** in the GUI or may be automatically surfaced in some circumstances. The operational data related to PLC **721** is shown in a tabular format in the present example. However, operational data may be presented in a variety of ways including tables, charts, graphs, raw data format, data pipelines, or similar.

(57) In addition to the presentation of operational data, the technology herein provides for access to data science tools, such that an automation engineer working in GUI **700** may perform analyses or use the operational data to extract information useful to their engineering tasks.

(58) In an exemplary embodiment, data table **740** is surfaced as a result of actions taken by wizard **730**. Wizard **730**, in some examples, employs one or more machine learning models to find data

relevant to control logic **720** and present the data in a way that may be useful or provide context for the automation engineer. Moreover, wizard **730** may, based on the operational data presented in data table **740**, make one or more recommendations to add or remove inputs or outputs from PLC **721**.

(59) FIG. **8** illustrates process **800** performed by a computing system for using an industrial programming wizard to assist writing or editing control logic for controlling industrial automation processes. Step **805** comprises, in a machine learning-based recommendation engine, generating a recommendation to add a component to the control logic based on an existing portion of the control logic. In generating the recommendation, the machine learning-based recommendation engine may employ one or more machine learning models to find that the component should be used or is typically used in similar environments. In step **810**, a notification component surfaces the recommendation to add the component to the control logic in a user interface of the programming environment. In response to noticing the recommendation, a user may choose to accept or decline the recommendation to add the component.

(60) In step **815**, in response to an acceptance of the recommendation, a programming component adds the component to the control logic. Once the component has been added to the control logic, it may require configuration in order to work properly in the intended environment. In step **820**, a configuration component configures the component based at least in part on the existing portion of the control logic. In some examples, configuring the component may include the configuration component and/or the machine learning-based recommendation engine guiding the user through configuration of the component.

(61) FIG. **9** illustrates process **900** performed by a computing system for performing variable relationship discovery to assist writing or editing control logic for controlling an industrial automation process. In step **905**, a machine learning-based analysis engine identifies a variable that is available to be utilized in the control logic for controlling an industrial automation environment. In some examples of step **905**, identifying a variable that is available to be utilized comprises finding that the variable that is heavily used in an associated industrial process but is not being used in the control logic. In other examples of step **905**, identifying the variable comprises accessing a control code repository and concluding that the variable is often used in the same or similar programs. In step **910**, the machine learning-based analysis engine determines that the variable is not utilized in the control logic. Alternatively, step **910** may comprise determining that the variable is under-utilized in the control logic. In step **915**, a recommendation component surfaces a recommendation to add the variable to the control logic. Process **900** may further include receiving an acceptance or rejection of the recommendation and adding the variable to the control logic.

(62) FIG. **10** illustrates process **1000** performed by a computing system for performing variable slimming to assist in editing control logic for controlling an industrial automation process. In step **1005**, a machine learning-based analysis engine performs an analysis of operational data from an industrial automation environment. In step **1010**, the machine learning-based analysis engine performs an analysis of the control logic for controlling the industrial automation environment. In step **1015**, the machine learning-based analysis engine identifies, based on the analysis of the operational data and the analysis of the control logic, a variable that is in the control logic but is not used in the operational data. In step **1020**, a notification component surfaces a notification that the variable is in the control logic but is not used in the operational data. In some examples, the notification that the variable is in the control logic but is not used in the operational data includes a recommendation to remove the variable from the control logic. Process **1000** may further include receiving an acceptance or rejection of a recommendation to remove the variable and removing the variable from the control logic.

(63) FIG. **11** illustrates process **1100** performed by a computing system for providing contextual data in an industrial programming environment associated with an industrial automation environment. In step **1105**, a user interface component displays the programming environment for

editing control logic associated with an industrial automation environment, wherein the operational data from the industrial automation environment is accessible from within the programming environment through a data pipeline. In step **1110**, a machine learning-based data science engine processes the operational data from the industrial automation environment to generate processed data for the industrial automation environment, wherein processed data may include any data that is formatted and presented in a useful way to a user of the programming environment. In step **1115**, the machine learning-based data science engine identifies a portion of the processed data relevant to a component of the control logic. In step **1120**, the user interface surfaces the portion of the processed data in the programming environment.

(64) FIG. **12** illustrates process **1200** performed by a computing system for providing contextual data in a data science environment associated with an industrial automation environment. In step **1205**, a user interface component displays a data science environment for analyzing operational data from an industrial automation environment, wherein control data from a programming environment associated with the industrial automation environment is accessible from within the data science environment through a data pipeline. In step **1210**, a machine learning-based context engine processes the control data from the programming environment to generate contextual data. In step **1215**, the user interface component surfaces a portion of the contextual data in the data science environment. In some examples, the contextual data comprises a model or similar rendering of the control code or a portion of the control code as it may be displayed in the programming environment.

(65) FIG. **13** illustrates computing system **1301** to perform machine learning assisted programming and data analytics according to an implementation of the present technology. Computing system **1301** is representative of any system or collection of systems with which the various operational architectures, processes, scenarios, and sequences disclosed herein for utilizing machine learning models within industrial programming and data science environments may be employed. Computing system **1301** may be implemented as a single apparatus, system, or device or may be implemented in a distributed manner as multiple apparatuses, systems, or devices. Computing system **1301** includes, but is not limited to, processing system **1302**, storage system **1303**, software **1305**, communication interface system **1307**, and user interface system **1309** (optional). Processing system **1302** is operatively coupled with storage system **1303**, communication interface system **1307**, and user interface system **1309**.

(66) Processing system **1302** loads and executes software **1305** from storage system **1303**. Software **1305** includes and implements machine learning assisted control programming **1306**, which is representative of any of the programming and analytic processes discussed with respect to the preceding Figures, including but not limited to industrial asset libraries, wizards, variable relationship discovery, and variable slimming. When executed by processing system **1302** to provide model implementation functions, software **1305** directs processing system **1302** to operate as described herein for at least the various processes, operational scenarios, and sequences discussed in the foregoing implementations. Computing system **1301** may optionally include additional devices, features, or functionality not discussed for purposes of brevity.

(67) Referring still to FIG. **13**, processing system **1302** may comprise a micro-processor and other circuitry that retrieves and executes software **1305** from storage system **1303**. Processing system **1302** may be implemented within a single processing device but may also be distributed across multiple processing devices or sub-systems that cooperate in executing program instructions. Examples of processing system **1302** include general purpose central processing units, graphical processing units, application specific processors, and logic devices, as well as any other type of processing device, combinations, or variations thereof.

(68) Storage system **1303** may comprise any computer readable storage media readable by processing system **1302** and capable of storing software **1305**. Storage system **1303** may include volatile and nonvolatile, removable and non-removable media implemented in any method or

technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of storage media include random access memory, read only memory, magnetic disks, optical disks, optical media, flash memory, virtual memory and non-virtual memory, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other suitable storage media. In no case is the computer readable storage media a propagated signal.

(69) In addition to computer readable storage media, in some implementations storage system **1303** may also include computer readable communication media over which at least some of software **1305** may be communicated internally or externally. Storage system **1303** may be implemented as a single storage device but may also be implemented across multiple storage devices or sub-systems co-located or distributed relative to each other. Storage system **1303** may comprise additional elements, such as a controller, capable of communicating with processing system **1302** or possibly other systems.

(70) Software **1305** (including machine learning assisted control programming **1306**) may be implemented in program instructions and among other functions may, when executed by processing system **1302**, direct processing system **1302** to operate as described with respect to the various operational scenarios, sequences, and processes illustrated herein. For example, software **1305** may include program instructions for performing variable slimming in control logic for industrial automation environments as described herein.

(71) In particular, the program instructions may include various components or modules that cooperate or otherwise interact to carry out the various processes and operational scenarios described herein. The various components or modules may be embodied in compiled or interpreted instructions, or in some other variation or combination of instructions. The various components or modules may be executed in a synchronous or asynchronous manner, serially or in parallel, in a single threaded environment or multi-threaded, or in accordance with any other suitable execution paradigm, variation, or combination thereof. Software **1305** may include additional processes, programs, or components, such as operating system software, virtualization software, or other application software. Software **1305** may also comprise firmware or some other form of machine-readable processing instructions executable by processing system **1302**.

(72) In general, software **1305** may, when loaded into processing system **1302** and executed, transform a suitable apparatus, system, or device (of which computing system **1301** is representative) overall from a general-purpose computing system into a special-purpose computing system customized to provide machine learning functionality to industrial programming and data science environments as described herein. Indeed, encoding software **1305** on storage system **1303** may transform the physical structure of storage system **1303**. The specific transformation of the physical structure may depend on various factors in different implementations of this description. Examples of such factors may include, but are not limited to, the technology used to implement the storage media of storage system **1303** and whether the computer-storage media are characterized as primary or secondary storage, as well as other factors.

(73) For example, if the computer readable storage media are implemented as semiconductor-based memory, software **1305** may transform the physical state of the semiconductor memory when the program instructions are encoded therein, such as by transforming the state of transistors, capacitors, or other discrete circuit elements constituting the semiconductor memory. A similar transformation may occur with respect to magnetic or optical media. Other transformations of physical media are possible without departing from the scope of the present description, with the foregoing examples provided only to facilitate the present discussion.

(74) Communication interface system **1307** may include communication connections and devices that allow for communication with other computing systems (not shown) over communication networks (not shown). Examples of connections and devices that together allow for inter-system communication may include network interface cards, antennas, power amplifiers, radiofrequency

circuitry, transceivers, and other communication circuitry. The connections and devices may communicate over communication media to exchange communications with other computing systems or networks of systems, such as metal, glass, air, or any other suitable communication media. The aforementioned media, connections, and devices are well known and need not be discussed at length here.

(75) Communication between computing system **1301** and other computing systems (not shown), may occur over a communication network or networks and in accordance with various communication protocols, combinations of protocols, or variations thereof. Examples include intranets, internets, the Internet, local area networks, wide area networks, wireless networks, wired networks, virtual networks, software defined networks, data center buses and backplanes, or any other type of network, combination of networks, or variation thereof. The aforementioned communication networks and protocols are well known and need not be discussed at length here.

(76) While some examples provided herein are described in the context of a firmware extension development or deployment device, it should be understood that the condition systems and methods described herein are not limited to such embodiments and may apply to a variety of other extension implementation environments and their associated systems. As will be appreciated by one skilled in the art, aspects of the present invention may be embodied as a system, method, computer program product, and other configurable systems. Accordingly, aspects of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects of the present invention may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(77) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or,” in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

(78) The phrases “in some embodiments,” “according to some embodiments,” “in the embodiments shown,” “in other embodiments,” and the like generally mean the particular feature, structure, or characteristic following the phrase is included in at least one implementation of the present technology, and may be included in more than one implementation. In addition, such phrases do not necessarily refer to the same embodiments or different embodiments.

(79) The above Detailed Description of examples of the technology is not intended to be exhaustive or to limit the technology to the precise form disclosed above. While specific examples for the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative implementations may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or subcombinations. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in

series, these processes or blocks may instead be performed or implemented in parallel or may be performed at different times. Further any specific numbers noted herein are only examples: alternative implementations may employ differing values or ranges.

(80) The teachings of the technology provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various examples described above can be combined to provide further implementations of the technology. Some alternative implementations of the technology may include not only additional elements to those implementations noted above, but also may include fewer elements.

(81) These and other changes can be made to the technology in light of the above Detailed Description. While the above description describes certain examples of the technology, and describes the best mode contemplated, no matter how detailed the above appears in text, the technology can be practiced in many ways. Details of the system may vary considerably in its specific implementation, while still being encompassed by the technology disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the technology should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the technology with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the technology to the specific examples disclosed in the specification, unless the above Detailed Description section explicitly defines such terms. Accordingly, the actual scope of the technology encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the technology under the claims.

(82) To reduce the number of claims, certain aspects of the technology are presented below in certain claim forms, but the applicant contemplates the various aspects of the technology in any number of claim forms. For example, while only one aspect of the technology is recited as a computer-readable medium claim, other aspects may likewise be embodied as a computer-readable medium claim, or in other forms, such as being embodied in a means-plus-function claim. Any claims intended to be treated under 35 U.S.C. § 112(f) will begin with the words “means for” but use of the term “for” in any other context is not intended to invoke treatment under 35 U.S.C. § 112(f). Accordingly, the applicant reserves the right to pursue additional claims after filing this application to pursue such additional claim forms, in either this application or in a continuing application.

Claims

1. A system for variable relationship discovery to assist in control logic editing, the system comprising: a memory that stores executable components; and a processor, operatively coupled to the memory, that executes the executable components, the executable components comprising: a machine learning-based analysis engine configured to, in a visual programming environment comprising a graphical user interface creating and editing the control logic comprising one or more iconic representations of industrial automation devices: obtain operational data generated during execution of an industrial process performed by a plurality of automated devices in an industrial automation environment; provide at least a portion of the operational data and the control logic as input to a machine learning model trained to identify variables in the operational data that are not utilized in the control logic; receive an output from the machine learning model, wherein the output identifies a variable in the operational data that is not included in the control logic; a recommendation component configured to, in the graphical user interface of the visual programming environment, suggest an auto-completion to add the variable to the control logic in response to the output identifying the variable, wherein the auto-completion comprises surfacing an iconic representation of the variable in relation to an iconic representation of a corresponding one of the industrial automation devices; and a programming component configured to: modify the

- control logic by adding the variable into the control logic and modifying the graphical user interface to include the iconic representation of the variable and a connection with the iconic representation of the corresponding one of the industrial automation devices in response to a user accepting the auto-completion, and transmit the modified control logic to a controller to configure the controller to perform the industrial process using the variable.
2. The system of claim 1, wherein the machine learning-based analysis engine is further configured to determine that the auto-completion to add the variable to the control logic should be surfaced based on a frequency of utilization of the variable in other control logic controlling other industrial automation environments of a same type as the industrial automation environment.
 3. The system of claim 2, wherein the frequency of utilization of the variable is determined based on information derived from one or more control code repositories to which the system is communicatively coupled.
 4. The system of claim 1, wherein the operational data is generated by the plurality of automated devices in the industrial automation environment.
 5. The system of claim 4, wherein: the variable is identified by the machine learning model as being correlative to changes in outcome in the operational data.
 6. A non-transitory computer-readable medium having stored thereon instructions that, in response to execution, cause a system comprising a processor to perform variable relationship discovery operations, the operations comprising: obtaining, in a visual programming environment comprising a graphical user interface creating and editing control logic comprising one or more iconic representations of industrial automation devices, operational data generated during execution of an industrial process performed by a plurality of automated devices in an industrial automation environment, wherein the control logic, when executed by a controller, instructs the plurality of automated devices to perform the industrial process; providing at least a portion of the operational data and the control logic as input to a machine learning model trained to identify variables in the operational data that are not utilized in the control logic; receiving an output from the machine learning model, wherein the output identifies a variable in the operational data that is not included in the control logic; suggesting, in the graphical user interface of the visual programming environment and in response to the output identifying the variable, an auto-completion to add the variable to the control logic, wherein the auto-completion comprises surfacing an iconic representation of the variable in relation to an iconic representation of a corresponding one of the industrial automation devices; modifying the control logic by adding the variable into the control logic and modifying the graphical user interface to include the iconic representation of the variable and a connection with the iconic representation of the corresponding one of the industrial automation devices in response to a user accepting the auto-completion; and transmitting the modified control logic to the controller to configure the controller to perform the industrial process using the variable.
 7. The non-transitory computer-readable medium of claim 6, the operations further comprising determining that the auto-completion to add the variable to the control logic should be surfaced based on a frequency of utilization of the variable in other control logic controlling other industrial automation environments of a same type as the industrial automation environment.
 8. The non-transitory computer-readable medium of claim 7, wherein the frequency of utilization of the variable is determined based on information derived from one or more control code repositories to which the system is communicatively coupled.
 9. The non-transitory computer-readable medium of claim 6, wherein the operational data is generated by the plurality of automated devices in the industrial automation environment.
 10. The non-transitory computer-readable medium of claim 9, wherein: the variable is identified by the machine learning model as being correlative to changes in outcome in the operational data.
 11. A method of recommending integration of salient variables in a visual industrial automation programming environment, the method comprising: obtaining, in the visual industrial automation

programming environment comprising a graphical user interface creating and editing control logic comprising one or more iconic representations of industrial automation devices, operational data generated during execution of an industrial process performed by a plurality of automated devices in an industrial automation environment, wherein the control logic, when executed by a controller, instructs the plurality of automated devices to perform the industrial process; providing, in the visual industrial automation programming environment, at least a portion of the operational data and the control logic as input to a machine learning model trained to identify variables in the operational data that are not utilized in the control logic; receiving in the visual industrial automation programming environment, an output from the machine learning model, wherein the output identifies a variable in the operational data that is not included in the control logic; suggesting, in the graphical user interface of the industrial automation visual programming environment, an auto-completion to add the variable to the control logic, wherein the auto-completion comprises surfacing an iconic representation of the variable in relation to an iconic representation of a corresponding one of the industrial automation devices; modifying, in the graphical user interface of the industrial automation visual programming environment, the control logic by adding the variable into the control logic and modifying the graphical user interface to include the iconic representation of the variable and a connection with the iconic representation of the corresponding one of the industrial automation devices in response to a user accepting the auto-completion; and transmitting the modified control logic to the controller to configure the controller to perform the industrial process using the variable.

12. The method of claim 11, further comprising determining that the auto-completion to add the variable to the control logic should be surfaced based on a frequency of utilization of the variable in other control logic controlling other industrial automation environments of a same type as the industrial automation environment.

13. The method of claim 12, wherein the frequency of utilization of the variable is determined based on information derived from one or more control code repositories to which the visual industrial automation programming environment is communicatively coupled.

14. The method of claim 11, wherein the operational data is generated by the plurality of automated devices in the industrial automation environment.

15. The method of claim 14, wherein: the variable is identified by the machine learning model as being correlative to changes in outcome in the operational data.
